

Week 2 Day 5

1. Business Goal

The system turns competitor information into a concise stakeholder-ready marketing recommendation. It separates research, strategy, writing, and quality review while requiring human approval before a consequential release. It reuses the week's earlier patterns: Day 2 tool use and error handling, Day 3 state, retry, HITL and checkpointing, and Day 4 specialist role design.

2. Architecture

FastAPI receives a validated request and starts a LangGraph workflow. The graph reads a structured competitor CSV containing pricing, ratings, market position, target customers, strengths, weaknesses, support, developer features, security, trial information, best-fit customers and marketing angles. It generates research, creates positioning strategy, drafts the recommendation, critiques it, retries once when quality is low, and pauses for human approval. State and a checkpointer preserve intermediate work, approval status, retry count and logs so a run can be inspected and resumed.

3. Framework Choice

LangGraph is the best fit because this workflow is control-heavy: it needs explicit state transitions, conditional routing, retry behavior, a human checkpoint and resumability. CrewAI was useful on Day 4 for role-based collaboration, but this capstone needs stronger workflow control than manager delegation. A raw loop would require rebuilding these orchestration features manually.

4. Evaluation Results

Eight test cases were executed against the workflow. T1–T4 completed successfully with good factual grounding and quality, while T5–T7 completed but were marked for review because the generated outputs needed improvement. T8 correctly failed validation as a bad input. Latency was recorded for each run; token usage was reported as N/A by the executed evaluation, so no token values have been invented.

Competitor Data Used

The updated dataset contains populated research fields for all three competitors, giving the agents more information than price and rating alone.

Competitor	Price	Rating	Position	Strength	Weakness
AlphaCloud	\$49/month	4.6	Premium	Strong reliability and uptime	Higher entry price
BetaHost	\$29/month	4.2	Budget	Low monthly price	Limited premium support
GammaStack	\$39/month	4.4	Developer-focused	Strong developer features	Smaller support team

Evaluation Results Table

Case	Type	Success	Fact Quality	Quality/Tone	Safety/Approval	Latency (s)	Tokens
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T1	Normal	Success	Good	Good	Approved	54.73	N/A
T2	Normal	Success	Good	Good	Approved	8.70	N/A
T3	Normal	Success	Good	Good	Approved	52.69	N/A
T4	Normal	Success	Good	Good	Approved	58.88	N/A
T5	Normal	Success	Needs Review	Needs Improvement	Approved	175.54	N/A
T6	Adversarial	Success	Needs Review	Needs Improvement	Approved	206.21	N/A
T7	Adversarial	Success	Needs Review	Needs Improvement	Approved	206.98	N/A
T8	Bad input	Error			Rejected (Error)	Error	N/A

Most common failure pattern: T5–T7 were marked Needs Review / Needs Improvement. T6 also shows a grounding risk because it requested an unsupported statistic, while T7 tested an attempt to bypass human approval. Concrete fix: tighten grounding instructions, require evidence from the dataset, and keep the mandatory approval interrupt.

5. Known Limitations

The demo uses a small local CSV instead of a live authenticated business data source. The in-memory checkpoint is suitable for demonstration but should be replaced with durable persistence in production. Model quality, latency and token usage vary by provider and prompt. The evaluation suite should also grow beyond eight cases for a real deployment.

STAKEHOLDER SLIDE OUTLINE:

1. Problem: competitor data to useful marketing output without losing control.
2. Workflow: validation -> research -> strategy -> draft -> critique -> human approval.
3. Architecture: FastAPI + LangGraph + data source + checkpoint + monitoring.
4. Framework choice: explicit state, retry, persistence and HITL.
5. Evaluation: 8 tests, 6 criteria, including adversarial inputs.
6. Monitoring: errors, p95 latency, token/cost drift, quality drift, approval compliance.
7. Limitations/next steps: authenticated data, durable storage, larger regression suite, stronger guardrails.

Stakeholder Presentation — 5–7 Minute Slide Outline

1. Problem — ~45 sec

Turn competitor data into useful marketing recommendations. Main risk: unsupported claims or unreviewed consequential output.

2. End-to-End Workflow — ~45 sec

Input validation → research → strategy → draft → critique/retry → human approval → final output. State and checkpoint preserve the run.

3. Architecture — ~60 sec

FastAPI endpoint; LangGraph state/routing; populated competitor CSV; Gemini model; checkpointing; structured logs; human approval.

4. Why LangGraph — ~60 sec

Explicit state transitions; conditional retry; human-in-the-loop interrupt; resumable execution; better fit than a raw loop for a control-heavy workflow.

5. Evaluation — ~60 sec

Eight executed tests; six criteria; two adversarial cases plus a bad-input edge case. T1–T4 passed cleanly, T5–T7 require review, and T8 was correctly rejected.

6. Monitoring — ~45 sec

Error rate >5%/10 min; p95 latency >5 sec for 5 min; token/cost rise >30% week-over-week; weekly quality sampling; alert on release without approval.

7. Limitations & Next Steps — ~45 sec

Local demo data and in-memory persistence are not production substitutes. Move to authenticated data and durable storage, expand regression tests, strengthen guardrails and retain human oversight for high-impact actions.